

OPTIMIZATION METHODS IN FINANCE: PROJECT REPORT

Robust Portfolio Risk Management under Distributional Uncertainty

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PART A: PAPER SUMMARY & ANALYSIS

1. Paper Details

Title: Multi-period CVaR-mean portfolio robust optimization model based on Wasserstein ambiguity sets

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2. Summary

Classical Markowitz mean-variance optimization operates on the assumption that future asset returns will precisely replicate historical probability distributions. This assumption systematically breaks down during market regime shifts, resulting in elevated out-of-sample drawdowns. This paper presents a **Distributionally Robust Optimization (DRO)** framework designed for strict capital preservation. Instead of optimizing against a single estimated distribution, the model optimizes against the worst-case distribution within a mathematically defined boundary - a "Wasserstein ambiguity set" - centered around empirical data. The objective is to minimize the worst-case **Conditional Value at Risk (CVaR)** over this space, yielding a portfolio explicitly hedged against distributional uncertainty and tail-risk events.

3. Model Formulation

The model translates portfolio choice into a tri-level Min-Max-Min risk-budgeting problem:

- **Inner Maximization:** The market adversarially selects the worst-case return distribution F_t from within a Wasserstein ball of radius ρ_t .
- **Constraint Layer:** The portfolio must satisfy a hard floor on expected return r_t under this worst-case distribution, remaining fully invested and long-only.
- **Outer Minimization:** The investor allocates weights x_t to minimize a convex combination of expected loss and CVaR tail risk.

$$\min_{\{x\}} \max_{\{F \in B_\rho\}} [(1 - \tau) E_F(-x^T) + \tau \text{CVaR}_\alpha(-x^T)]$$

4. Methodology Explanation

- **Lagrangian Duality Reformulation:** Exploiting the piecewise-linear structure of the CVaR loss alongside the dual representation of the 1-Wasserstein metric, the authors collapse the highly complex tri-level problem into a tractable Linear Program (LP), allowing deterministic solutions without relying on heuristic approximations.
- **Stationary Block Bootstrapping (SBB):** Because independent random sampling destroys financial time-series momentum, SBB is used to draw contiguous blocks of historical data. This preserves serial dependence and volatility clustering when generating robust stress scenarios to form the ambiguity set.
- **Dynamic Regime Adaptation:** An EWMA volatility z-score is passed through a bounded sigmoid function to dynamically adjust the risk-aversion parameter (τ), naturally tightening the risk budget during high-volatility market crashes.

PART B: IMPLEMENTATION ON DATASET

5. Data & Implementation Configuration

The methodology was translated into Python utilizing `scipy.optimize`. It was applied to the provided dataset of 30 Indian equities (Data for OMF.xlsx). To rigorously test the paper's claims regarding out-of-sample robustness, an 80/20 chronological train-test split was executed.

Three models were constructed and compared using an identical target-return mandate:

1. **Markowitz Max-Sharpe:** Classical mean-variance tangency portfolio (Non-robust benchmark).
2. **Historical CVaR-Optimal:** Rockafellar-Uryasev LP solved strictly on the raw historical scenarios.
3. **Wasserstein-Robust CVaR:** The proposed LP solved across a Stationary Block Bootstrapped scenario pool (500 paths) to explicitly model distributional uncertainty.

6. Results (In-Sample vs. Out-of-Sample)

Note: Models were trained on the first 80% of data. The Adaptive Risk-Return Weight (τ) resolved to 0.4439. Weights were then frozen and evaluated on the unseen final 20%.

Period	Portfolio Model	Ann. Return	Ann. Volatility	Sharpe	95% CVaR
In-Sample	Markowitz Max-Sharpe	26.87%	13.92%	1.93	29.07%
In-Sample	Historical CVaR	15.03%	11.44%	1.31	22.69%
In-Sample	Wasserstein-Robust CVaR	15.70%	11.47%	1.37	23.16%
Out-of-Sample	Markowitz (Frozen)	2.02%	12.54%	0.16	30.25%
Out-of-Sample	Historical CVaR (Frozen)	1.98%	11.26%	0.18	25.68%
Out-of-Sample	Wasserstein-Robust CVaR	2.47%	11.11%	0.22	25.09%

7. Interpretation & Conclusion

The empirical results directly validate the paper's thesis regarding out-of-sample robustness. On historical training data, the classical Markowitz portfolio posts an exceptional Sharpe ratio (1.93), but achieves this by over-fitting to the in-sample covariance, retaining a massive 29.07% tail risk (CVaR). When exposed to unseen out-of-sample data, the Markowitz portfolio's performance collapses to a 0.16 Sharpe ratio, and its tail risk worsens to 30.25%.

In contrast, explicitly budgeting for distributional uncertainty via the **Wasserstein-Robust CVaR model** required a known trade-off during training (an in-sample Sharpe of 1.37). However, it proved superior during the out-of-sample stress test, delivering the highest risk-adjusted return (0.22 Sharpe) among the models, while successfully maintaining the lowest realized tail loss magnitude (25.09% CVaR). For institutional portfolio construction, this confirms that distributionally robust optimization offers effective downside protection and mitigates performance degradation when future market conditions deviate from historical norms.